Chapter 11, Problem 92CP

Problem

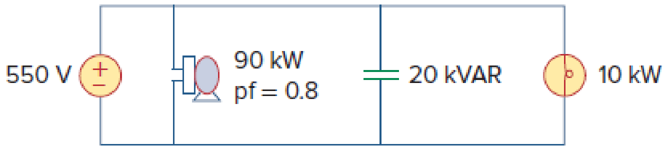
As shown in Fig. 11.97, a 550-V feeder line supplies an industrial plant consisting of a motor drawing 90 kW at 0.8 pf (inductive), a capacitor with a rating of 20 kVAR, and lighting drawing 10 kW.

(a) Calculate the total reactive power and apparent power absorbed by the plant.

(b) Determine the overall pf.

(c) Find the magnitude of the current in the feeder line.

Figure 11.97:



Step-by-step solution

Step 1 of 7

Refer to Figure 11.97 in the text book.

Calculate the value of the power factor angle at the motor.

$$\begin{aligned}\theta &= \cos^{-1}(0.8)\\ &= 36.87^\circ\end{aligned}$$

Therefore, the value of the power factor angle at the motor is 36.87°.

Step 2 of 7

Calculate the apparent power absorbed by the motor.

$$S = \frac{P}{\text{pf}}$$

Substitute 90 kW for P and 0.8 for pf.

$$\begin{aligned}S &= \frac{90\text{ k}}{0.8}\\ &= 112.5\text{ kVA}\end{aligned}$$

Therefore, the value of the apparent power absorbed by the motor is 112.5 kVA.

Step 3 of 7

Calculate the value of the reactive power absorbed y the motor.

$$Q = S\sin\theta$$

Substitute 112.5 kVA for S and 36.87° for θ.

$$\begin{aligned}Q &= (112.5\text{ k})\sin(36.87^\circ)\\ &= (112.5\text{ k})(0.6)\\ &= 67.5\text{ kVAR}\end{aligned}$$

Therefore, the value of the reactive power absorbed by the motor is 67.5 kVAR.

Step 4 of 7

(a)

Calculate the total complex power absorbed by the load.

$$S = 90\text{ k} + j67.5\text{ k} + j20\text{ k} + 10\text{ k}$$
$$= (100\text{ k} + j87.5\text{ k})\text{ VA}$$

Therefore, the value of the total complex power absorbed by the load is,

$(100\text{ k} + j87.5\text{ k})\text{ VA}$.

Compare $(100\text{ k} + j87.5\text{ k})\text{ VA}$ to $P_T + jQ_T$, where P_T and Q_T represents the total real and total reactive power absorbed by the load respectively.

Therefore, the values of P_T and Q_T are 100 kW and $\boxed{87.5\text{ kVAR}}$ respectively.

Step 5 of 7

Calculate the value of the total apparent power S_T.

$$S = \sqrt{P^2 + Q^2}$$

Substitute 100 kW for P and 87.5 kVAR for Q.

$$S_T = \sqrt{(100\text{ k})^2 + (87.5\text{ k})^2}$$
$$= \sqrt{10000 + 7656.25}\text{ k}$$
$$= \sqrt{17656.25}\text{ k}$$
$$= 132.88\text{ kVA}$$

Therefore, the value of the total apparent power is $\boxed{132.88\text{ kVA}}$.

Step 6 of 7

(b)

Calculate the value of the overall power factor.

$$\text{pf} = \frac{P}{S}$$

Substitute 132.88 kVA for S and 100 kW for P.

$$\text{pf} = \frac{100\text{ k}}{132.88\text{ k}}$$
$$= 0.7526$$

Therefore, the value of the overall power factor is $\boxed{0.7526}$.

Step 7 of 7

(c)

Calculate the magnitude of current in the feeder.

$$I = \frac{S_T}{|V|}$$

Substitute $132.88\ \text{kVA}$ for S and $550\ \text{V}$ for $|V|$.

$$I = \frac{132.88\ \text{k}}{550}$$
$$= 241.6\ \text{A}$$

Therefore, the magnitude value of the current in the feeder is $\boxed{241.6\ \text{A}}$.